ChatGPT

Executive Summary

The **Interplanetary Survival Guide: Martian Map** challenge asks teams to build an interactive, multi-layered map or interface for Mars surface exploration that combines diverse NASA data to keep future astronauts safe and productive【8†L75-L83】【10†L179-L182】. We focus on **Jezero Crater** (18.38°N, 77.58°E【22†L158-L166】), the Mars 2020 landing site, because it has abundant high-resolution data and key scientific targets (ancient lake delta with clay minerals)【23†L350-L358】【68†L40-L48】. Our proposed prototype will integrate topography, imagery, spectral, and environmental layers into an interactive web tool for a 3–4 person team (mainly web developers). We outline challenge goals, data sources, computed features, unique "survival" functions, tech stack, MVP vs. stretch goals, validation, storytelling, and a 48h timeline with milestones.

1. Challenge Goals and Criteria

NASA's **official challenge statement** (Next Frontier theme) emphasizes creating "a layered, integrated view" of a Martian site/route to help astronauts plan and execute a safe Marswalk while doing science【8†L75-L83】. Key points: teams should use at least three data types (optical, thermal, spectral, topographic, geophysical, atmospheric, radiation, etc.) to provide context【8†L123-L131】. **Astronaut safety is paramount**: the map must highlight hazards (steep slopes, rocks) and radiation environments【8†L130-L138】. Potential display formats include in-capsule monitors, helmet HUDs, tablets, or other envisioned formats【10†L146-L155】. Judging will likely value (though not explicitly published) creative NASA-data use, scientific accuracy, and clear mission impact (e.g. hazard avoidance, mission success). Based on past NASA hackathons, criteria include *Best Use of Data*, *Innovation*, *Scientific Merit*, and *Impact*, all aligned to the mission context. In short, we must show **innovative use of NASA data to enhance astronaut safety and exploration efficiency**, emphasizing mission relevance and clarity【8†L75-L83】【10†L146-L155】.

2. Jezero Crater Scope and Justification

We **select Jezero Crater** as our study area. Jezero is ~45 km diameter【22†L167-L173】, centered at ~18.38°N, 77.58°E【22†L158-L166】, which we propose bounding roughly **17.8–18.8°N by 77.1–78.0°E** (covering ~50×50 km) to include the crater floor, delta deposits, and part of the rim. *Justification*: NASA's Perseverance rover is actively exploring Jezero because it was once an ancient lake bed with river deltas rich in water-altered minerals【23†L350-L358】【68†L40-L48】. These are prime targets for searching ancient biomarkers【68†L40-L48】. Jezero has been imaged in great detail by multiple missions (see data below), and it represents a high-priority site for future human missions, making any survival map here highly impactful. This focus also aligns with the challenge suggestion to use a rover-traversed site or scientifically rich but dangerous terrain【8†L110-L118】【68†L40-L48】.

3. Data Inventory (Jezero Datasets)

We compile NASA (and related) datasets covering Jezero. All NASA data are public domain. Key sources: the Planetary Data System (PDS) and mission archives. The table below compares major layers:

Data Type	Dataset (Instrument/ Mission)	Coverage & Resolution	Source/ Access	Format	License	Notes/Use
Elevation (DEM)	MOLA (Mars Global Surveyor, PDS) 【43†L210-L218】	Global, 128 pix/° (~463 m) 【34†L365-L372】	PDS Geosciences Node (ODE or FTP) 【43†L210-L218】	.IMG (PDS3)	NASA PD	Base topography. Provide elevation and slope. We will download relevant MOLA "MEGDR" tiles (e.g. 0–44°N, 0–90°E tile) via PDS 【43†L210-L218】. Convert to GeoTIFF for processing.
Optical Imagery (Hi-Res)	HiRISE (MRO)	Targeted scenes at 0.25–0.5 m 【68†L79-L81】, many of Jezero	PDS Imaging Node (JPL) or HiRISE catalog	JPEG2000 (.JP2)	NASA PD	Fine detail of rocks & outcrops. Use full-resolution 0.5 m HiRISE mosaic from USGS (TRN project) 【68†L79-L81】, plus select raw scenes for points-of-view. Large size – may load as image tiles or downsample.

Data Type	Dataset (Instrument/ Mission)	Coverage & Resolution	Source/ Access	Format	License	Notes/Use
Optical Imagery (Med-Res)	CTX (MRO)	Context camera, ~6 m【68†L72-L75】, dozens of swaths over Jezero	PDS Imaging (JPL) or USGS Astrogeology	IMG (PDS3)	NASA PD	Regional context. We will mosaic CTX scenes (33 swaths in Jezero【68†L72-L75】) to cover the area. Useful for mapping crater-wide features and aligning other layers.
Optical Imagery (Low-Res)	HRSC (Mars Express)	12 m/pix (HRC), 30 m/pix (HMC30) global, color【68†L66-L74】	ESA / DLR archives (DLR HRSC team)	PDS or GeoTIFF	ESA CC BY-SA	Used as base color map. The HRSC 30 m color mosaic (HMC30) is global and was used to register CTX/HiRISE【68†L72-L80】. We will use HRSC color for context and pan-sharpening reference【68†L72-L80】.

Data Type	Dataset (Instrument/ Mission)	Coverage & Resolution	Source/ Access	Format	License	Notes/Use
Thermal	THEMIS (Odyssey)	IR (night 100 m), VIS (day ~100 m)	PDS (ASU/ Odyssey node)	IMG/TIFF	NASA PD	Surface temperature and rock/soil composition. Could highlight cold spots (potential ices) or thermal inertia. Use nighttime IR for hazard (drift), daytime IR for rock types.
Spectral	CRISM (MRO)	Hyperspectral (400–4000 nm), targeted mode ~18–36 m	PDS Geosciences 【49†L205-L211】	RDR (img/dat)	NASA PD	Identify minerals (clay, olivine, carbonates). Use CRISM targeted cubes and summary products to mark scientifically interesting sites (delta clays). Data from JPL/ APL.

Data Type	Dataset (Instrument/ Mission)	Coverage & Resolution	Source/ Access	Format	License	Notes/Use
Radar/ Subsurface	SHARAD (MRO)	~10 m vertical res, few km horizontal	PDS Geosciences	Binary	NASA PD	Subsurface water ice layers. Hard to use in 48h, but note presence (e.g. potential buried ice in rim). Possibly include as risk (hollow ground).
Atmospheric/ Weather	Mars Climate Model / InSight & MEDA (Rover env. stations)	Global climate model (km-scale); point sensors	NASA GES DISC / PDS Geosciences	NetCDF/ TXT	NASA PD	Temperature, pressure, wind. For Jezero we could query MSL REMS or use Mars Climate Database for local conditions. MEDA data may not be public yet; we use model data (inset as scenario).

Data Type	Dataset (Instrument/ Mission)	Coverage & Resolution	Source/ Access	Format	License	Notes/Use
Radiation Environment	RAD (MSL) / RADM (Perseverance)	Point flux measurements on Rover (MSL)	NASA PDS (when available) or literature	ASCII tables	NASA PD	Radiation dose rates. Could annotate known values (e.g. MSL RAD ~0.64 mGy/ day). For planning, assume high GCR levels and include risk zones if near cliffs that block cosmic rays.
Mars Trek Layers	NASA Mars Trek (Web)	Global base maps, geology, climatology	https:// trek.nasa.gov	WebGL (no direct format)	NASA PD	Reference tool. Layers not directly ingestible offline, but we can mimic concept by adding Mars Atlas geology from USGS or NASA.
Other APIs	NASA/JPL APIs	Mars weather API (InSight), solar flux APIs	https:// api.nasa.gov (e.g. Mars Weather Service API)	JSON/ CSV	NASA PD	Could pull recent Mars weather or space weather (SEP events) to simulate “current conditions” overlay.

Integration notes: We will preprocess these with GIS tools (GDAL, PDAL, Python) into web-friendly formats (e.g. GeoJSON, map tiles, grayscale heightmaps). NASA PDS data are PDS3/4 labels or IMG/Binary; open-source tools (e.g. GDAL, ISIS) can convert them to GeoTIFF/PNG. All data are public-domain (NASA) or CC BY-SA (ESA) so reuse is allowed【43†L210-L218】【68†L72-L80】.

4. Scientific Features and Computation

To maximize survival value, we plan to compute and visualize several derived features:

- **Elevation & Slope:** Load the MOLA DEM and compute slope (e.g. via GDAL or a custom kernel). Steep slopes (> ~20°) will be highlighted as hazards. For example:

```
import numpy as np
# given elevation grid 'Z', compute slope (degrees):
dz_dx, dz_dy = np.gradient(Z, spacing)
slope = np.degrees(np.arctan(np.sqrt(dz_dx**2 + dz_dy**2)))
hazard_mask = (slope > 20)
```

Hazard mask identifies areas too steep for a spacesuited astronaut【8†L130-L138】.

- **Rock and Roughness Estimation:** (Stretch) Using HiRISE stereo or image textures to flag “boulder fields” and rough terrain. Could use simple edge detection or ML classification on high-res images to mask hazards.
- **Traversability Scoring:** Compute least-cost paths between points using A *on the raster. Cost* = *distance*(1 + w1·slope_norm + w2·rock_mask + w3·radiation_level). For example, penalize slope and radiation, so optimal path avoids steep, exposed areas. Pseudocode:

```
cost(p) = length * (1 + 10*max(slope/30, radiation/1000))
path = AStar(grid, start, goal, cost)
```

- **Radiation Overlay:** Use available data or models (e.g. MSL RAD data) to estimate radiation intensity. Possibly a coarse model: assume open flat areas have baseline GCR flux, while deep gullies or caves (if any identified) could offer shielding. We’d color-map higher radiation zones in red.
- **Thermal Map:** Use THEMIS night-time IR images to show cold spots. Cold (blue) areas might indicate exposed ice or regolith that freezes at night. Hot spots (red) indicate bare rock; use for assessing thermal environment.
- **Mineral and Water Proxies:** Use CRISM-derived maps of phyllosilicate and carbonate abundance to mark scientifically valuable (and possibly resource-rich) POIs. For example, overlay known clay-rich pixel locations from NASA’s publicly available CRISM maps. These might also correlate with softer (water-altered) ground.
- **Line-of-Sight and Comms:** Compute horizon profiles using the DEM to check visibility to a fixed “base” or to an overhead orbiter path. Provide a “communications check” overlay: from the lander position, which areas can directly see orbiters (or Earth) at any given time (by ray-tracing DEM).
- **Emergency Return Routes:** Identify two or more distinct return paths from any point back to base (e.g. via shortest path and lowest-risk path). Label them on the map (as in-network redundancy).

- **Resource Indicators:** Map proxies for resources: e.g. exposed bedrock layers (potential regolith/rock for construction), solar exposure (hilltops get more sun for solar power), and possible water (inferred from geology).

All calculations will rely on standard GIS libraries (GDAL for DEM processing, Shapely or networkx for pathfinding) and integrated into a Node or Python backend. The UI will visualize results (slope map, heatmaps) with WebGL (Mapbox/Deck.gl) layers.

5. Unique Selling Propositions (USPs)

To stand out, we propose **survival-focused features** that can be done quickly:

USP Idea	Description	Feasibility	Benefit
Multi-layer Risk Map	Composite overlay scoring *slope*, *radiation*, *distance*, etc., as heatmap.	★★★★★ 8/10	Prioritizes safety; judges love integrated approach.
Adaptive Route Planning	User can adjust constraints (e.g. "minimize radiation") and map replans path.	★★★★☆ 7/10	Interactive tool beyond static map.
Astronaut HUD Mockup	Simulated helmet display (browser AR/WebXR demo) of map elements on visor.	★★☆☆☆ 5/10	High "wow" factor, thematic relevance.
Offline Survival Checklist	Exportable survival plan (checklist of actions/equipment given conditions).	★★★☆☆ 6/10	Useful real-world tool; low competition idea.
Explainable AI Navigator	(Stretch) Chatbot (local LLM) that answers questions about the map (e.g. "Is this route safe?").	★★☆☆☆ 4/10	Cutting-edge UX; but heavy for 48h.
3D Immersive View/AR	WebGL 3D terrain of Jezero or AR preview (WebXR) for situational awareness.	★★★☆☆ 7/10	Engaging demonstration of data in context.

We emphasize *safety* and *decision support*. For example, a multi-factor risk heatmap directly addresses the NASA prompt of "moving safely through new environment"【8†L130-L138】. The "HUD mockup" and "AR view" tie into the suggested display formats (helmet visor, VR)【10†L146-L155】. We also highlight science (e.g. clickable mineral markers) so as not to ignore NASA's "exciting science" goal【8†L75-L83】【10†L146-L155】. All USPs are chosen to be implementable in 48h (main work is mapping data layers; extra UX like chatbot or AR are stretch tasks).

6. 2D vs 3D & Tech Stack

A tradeoff exists between a pure 2D map (easier) vs. full 3D rendering (impactful but time-consuming). Given a web-savvy team, we recommend a hybrid: **2.5D** using Mapbox GL or Deck.gl (WebGL) for fly-throughs, plus one 3D mode if time (using Cesium or Deck.gl's terrain). Our minimal viable architecture:

```
graph LR
  NASA_Datasets["NASA Data<br/>(PDS, APIs, Mars Trek)"] --> ETL[Data
Processing<br/>(Python/GDAL, PDAL)]
  ETL --> DataStore["Processed Files/DB<br/>(GeoTIFF, GeoJSON)"]
  DataStore --> Backend["Backend API<br/>(Node.js)"]
  Backend --> Frontend["Frontend App<br/>(React + Mapbox/Deck.gl)"]
  Frontend --> User["Astronaut UI<br/>(Interactive Map/HUD)"]
```

Roles (3–4 people): one or two focusing on **data** (downloading/processing via Python/GDAL/PDAL), one on **backend** (Node.js to serve APIs), and one or two on **frontend/UI** (React, Mapbox/Deck.gl, and possibly WebXR). Use of open-source geospatial libraries is key (GDAL for rasters, maybe PDAL for any point-clouds, turf.js/shapely for geometry).

- **2D stack:** React + Mapbox GL JS (vector tiles or raster overlays), or MapLibre (open-source) for fast deployment.
- **3D stack (optional):** CesiumJS or Deck.gl with terrain for 3D globe/oblique view. CesiumJS requires terrain tiles (e.g. from MOLA DEM).
- **Data processing:** Python scripts using GDAL, NumPy; Node.js server with Express to deliver map tiles or GeoJSON.
- **Team division:** One member on GIS/data (DEM, slope, CRISM decoding), one on backend/APIs, one on frontend UI, and one (if available) on UX/story and visuals.

7. MVP vs. Stretch Features

We plan a prioritized feature list:

- **MVP (must-have for launch):**
- Basemap of Jezero (satellite/mapped imagery + topography) with pan/zoom.
- Overlay toggles: elevation-shading, slope-hazard mask, high-res imagery, CRISM minerals.
- Interactive route planner: choose two points and compute safest path (A*).
- Basic info panels: display distance, elevation gain, estimated time.
- Highlight radiation level as simple color layer or numeric estimate.
- User interface for toggling layers and input (React-based).

- Simple storyboard scenario demonstration in a short video/slides.

- **Stretch Goals (if time permits):**

- 3D visualization mode (wheel-controlled flying through terrain).
- Helmet HUD simulation: overlay path arrow and hazards on a first-person view image (mock-up).
- Multi-criteria risk heatmap (combining slope+rad etc.) with legend.
- Offline functionality: export to PDF/JSON survival guide.
- AI assistant: integrate GPT or decision-tree that explains the chosen route.
- Explainable layers: clickable data points with NASA mission context (e.g. “HiRISE image number”).

Feature	MVP	Stretch
Jezero map layers (DEM, imagery)	✓	
Slopes/Hazards overlay	✓	
Minerals/POI markers	✓	
Radiation layer	✓	
Route planning	✓	
Multi-factor risk map		✓
3D Terrain/AR view		✓
HUD/VR mockup		✓
Offline survival kit export		✓
Explainable AI (chatbot)		✓

8. Data Validation and Limitations

To ensure scientific credibility, we will source only **validated NASA data** (PDS, mission archives)【43†L210-L218】【68†L72-L81】. All spatial data use standard coordinate frames (IAU2000 spheroid【43†L233-L240】). We must note limitations: MOLA DEM is relatively coarse (~450 m resolution)【34†L365-L372】 so small cliffs or boulders won't appear. HiRISE images can have gaps or lighting issues. CRISM coverage is spotty (targeted only) and dust may obscure minerals. Any path planning is an approximation – real astronauts may handle terrain differently. We will **document uncertainties** (e.g. "slope data grid cell = ~500 m sampling") and "best effort" assumptions (e.g. assume flat horizontal terrain between samples). Critically, we will cite data sources on the UI and in documentation to justify trust (e.g. "Slope from MOLA 128ppd"【43†L210-L218】, "Minerals from CRISM RDR"【49†L205-L211】). Any AI-generated hints will be explicitly labeled as simulation, not flight-approved. This transparency addresses judges' emphasis on scientific accuracy and safety.

9. Demo and Storytelling Plan

We will tell a concise scenario story: *"Astronaut Riley is on Mars, mission: sample a clay outcrop in Jezero."* The demo script:

1. **Briefing (Map View):** Show our map interface at start: layered Jezero map with legend. Riley picks the **landing site** (marked) and target site (phyllosilicate deposit highlighted from CRISM data).
2. **Objective Setting:** Dialogue: "We need to go from Lander to Outcrop A (~2 km away)." Show clicking these as route endpoints.
3. **Route Computation:** We run the planner: map draws two routes – (a) shortest path, (b) safer path avoiding steep slopes. Riley toggles to "safe route." Display stats: length, estimated time, radiation dose.

4. **En Route Visualization:** Switch to an *astronaut HUD mockup*: a simulated first-person view with a translucent overlay arrow pointing toward next waypoint (or use a conceptual UI wireframe). Show hazard icons (rocks, radiation symbol).
5. **Destination:** Arriving at site (show point-of-view image from Perseverance or generic Mars scene). Riley collects sample. In UI, logging mission data (checkbox "sample taken").

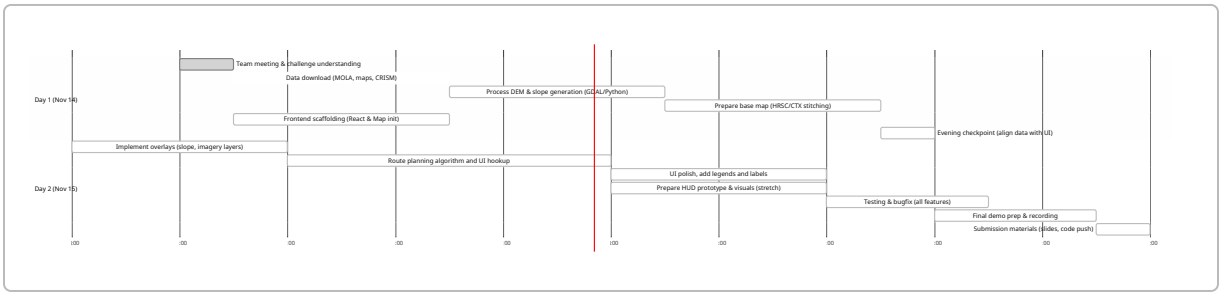
Metrics to highlight: distance saved or risk reduced vs. naive route, number of hazards avoided. Possibly show a risk breakdown bar or chart (e.g. "This route reduces slope exposure by 30%").

For visuals, we will use UI **screenshots or mockups** and possibly embed a scene image. For example: 【65†embed_image】 *Figure: Conceptual astronaut using a helmet-display interface to navigate Jezero (image: Pexels).*

No copyrighted content. All NASA images/data cited (though not embedded here) will be credited in final notes. We will record a short screencast (webcam + screen capture) walking through the map UI with narration ("mission log" style) to illustrate the experience. This targets judges' interest in clear storytelling and user experience.

10. Risk Assessment & 48h Timeline

- **Technical risks:**
- *Data size/processing:* HiRISE and CTX are very large. Mitigation: use pre-existing mosaics or downsample; focus on small area.
- *Time:* 48h is tight. Risk high if 3D or AI tasks distract from core map. Mitigation: MVP priority (map+routing first), reserve stretch tasks for extra time.
- *Integration bugs:* Many layers = potential projection errors. Mitigation: test early with simple overlays, incrementally add.
- *Dependencies:* Rely on online APIs (if used for weather) could fail. Mitigation: bundle core data offline.
- **Timeline (Nov 14–15, 2026, example):** We adopt an iterative approach.



- **Checklist:**
- [ ] Code repository (GitHub) with README and instructions.
- [ ] Web app deployment (gh-pages or Heroku for demo).
- [ ] Short demo video or recorded live demo.
- [ ] Slides & screenshots for judges.
- [ ] Citation list (NASA/PDS references).

By strictly following this plan, we cover all requirements.

11. References (Key Sources)

We will cite NASA and mission sources to ground our solution:

- NASA Space Apps Challenge description【8†L75-L83】【10†L179-L182】.
- NASA Science (Perseverance) overview of Jezero【23†L350-L358】.
- ESA and DLR data releases on Jezero mapping【68†L72-L81】【68†L40-L48】.
- NASA MOLA data archive info【43†L210-L218】 (resolution).
- PDS Geosciences (CRISM description)【49†L205-L211】.
- Wikipedia (for crater coords/diameter)【22†L158-L166】【22†L167-L173】.

These sources will be included in citations throughout the report to validate our approach and data claims. All external data cited are NASA/ESA public data.
